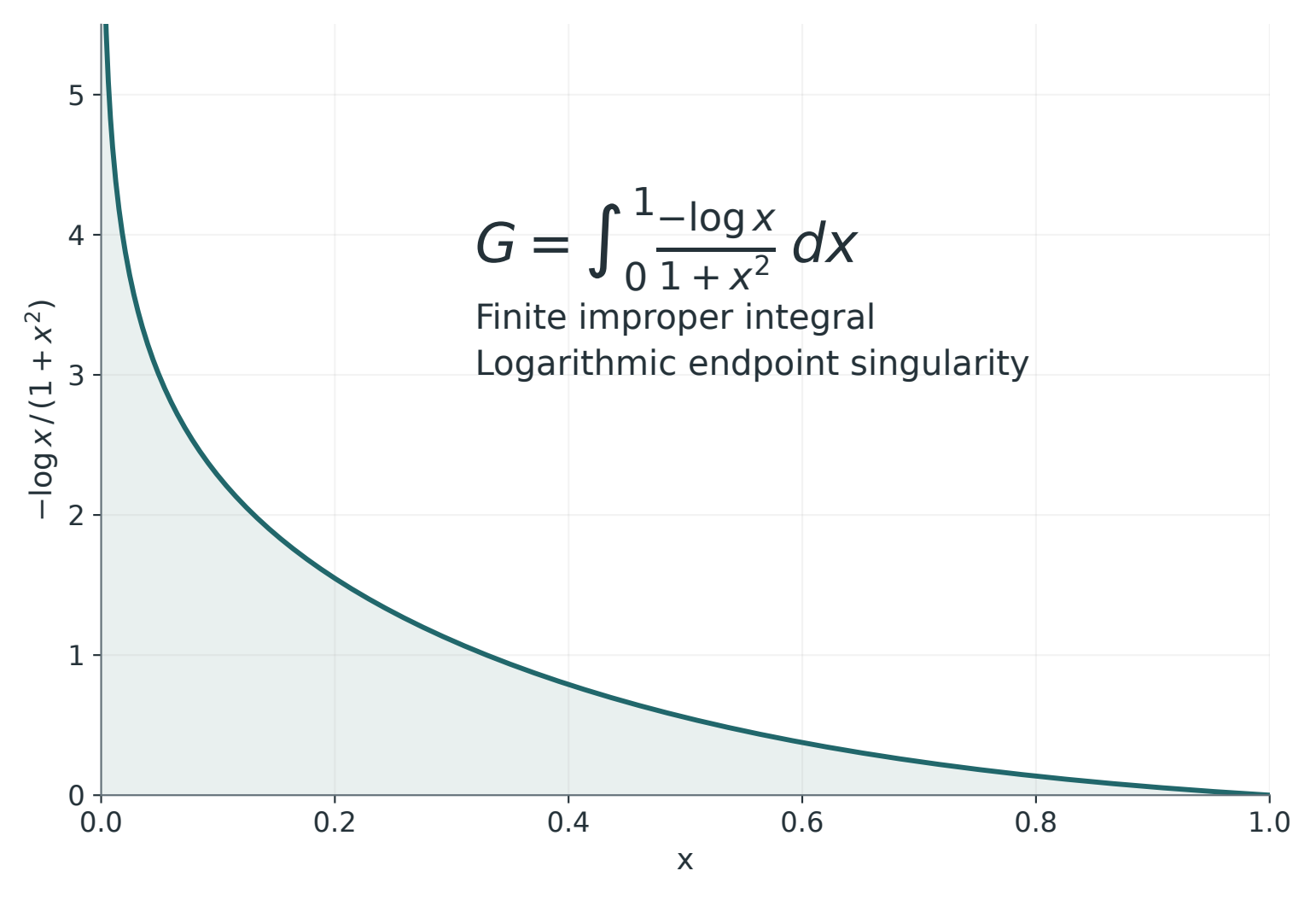
$G = \int_0^1 \frac{-\log x}{1+x^2}\,dx$
Finite improper integral
Logarithmic endpoint singularity
$-\log x/(1+x^2)$
0
1
2
3
4
5
0.0
0.2
0.4
0.6
0.8
1.0
x